

SHAHILULLA S

SOFTWARE DEVELOPMENT ENGINEER | BACKEND & DISTRIBUTED SYSTEMS

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PROFESSIONAL SUMMARY

Proactive Integrated M.Tech Computer Science & Engineering student with strong fundamentals in **Data Structures & Algorithms (DSA)**, **Object-Oriented Programming (OOP)**, and **Database Management Systems**. Proven hands-on experience designing scalable backend architectures, distributed data pipelines, and optimized relational databases using **Java, Python, SQL, and PostgreSQL**. Experienced in developing microservices, tuning query performance, and implementing robust automated validation workflows.

TECHNICAL SKILLS

Programming Languages:	Java, Python, SQL, C/C++, PL/pgSQL
Core CS Fundamentals:	Data Structures & Algorithms (DSA), Object-Oriented Design (OOP), DBMS, Operating Systems, Computer Networks
Backend & Databases:	PostgreSQL, MySQL, Database Architecture, Query Optimization, Indexing, ACID Transactions, REST APIs
Distributed Systems & Cloud:	Apache Spark, PySpark, Distributed Data Pipelines, Microservices, AWS (EC2, S3), GCP, Databricks
DevOps & Tools:	Docker, Git, GitHub, Linux/Unix, ETL Workflows, Power BI, Excel

SOFTWARE ENGINEERING PROJECTS

Adaptive AI-Enhanced Root Cause Analysis Framework | Research Project

Cloud-Native Systems

Technologies: Python, Microservices, Distributed Observability, Cloud-Native Architecture, Docker

- Researched and developed an adaptive, explainable root-cause analysis engine for cloud-native microservices under partial observability.
- Architected dependency graph algorithms and telemetry tracing to analyze failure propagation across distributed service endpoints.
- Engineered interpretable inference models that generate automated diagnostic reasoning, significantly reducing system debugging latency.

NexusFlow — Modular Backend & Analytics Platform

Scalable Systems

Technologies: PostgreSQL, SQL, Data Modeling, Python/Java, RESTful Services, ETL Architecture

- Architected an end-to-end backend platform and analytics engine handling multi-tenant library and educational data workloads.
- Designed normalized relational database schemas with custom indexing strategies, boosting analytical query throughput by 25%.
- Implemented modular REST APIs and automated data pipelines with comprehensive validation layers to ensure transactional consistency.

Healthcare Distributed Processing Engine & Pipeline

Distributed Big Data

Technologies: Python, PySpark, PostgreSQL, GCP, Databricks, Cloud-Native Architecture

- Constructed a distributed data processing engine in PySpark on GCP/Databricks to handle high-volume healthcare datasets.
- Implemented automated validation, anomaly sanitization, and schema enforcement algorithms, achieving 99.9% data reliability.
- Optimized distributed transformations and memory partitions, reducing cloud compute resource consumption by 15%.

Library Management System & Transaction Backend

Database Engineering

Technologies: PostgreSQL, Relational Constraints, PL/pgSQL, Triggers & Procedures, ACID Compliance

- Engineered an enterprise relational database backend enforcing ACID compliance and complex relational integrity constraints.
- Developed modular PL/pgSQL triggers, stored procedures, and functions for automated overdue calculations and concurrency-safe transactions.
- Structured indexed views and analytical queries, accelerating operational reporting and checkout efficiency by 30%.

EDUCATION

Integrated M.Tech in Computer Science & Engineering

2022 – 2027

Erode Sengunthar Engineering College, Tamil Nadu | Coursework: DSA, OOP, DBMS, OS, Networks

CGPA: 7.8 / 10.0

KEY ACHIEVEMENTS & TECHNICAL IMPACT

- Backend Optimization:** Enhanced query performance and data workflows by 30% via schema indexing and stored procedures.
- Analytics Speed:** Boosted platform analytics response times by 25% over 6 months of iterative architectural improvements.
- Resource Efficiency:** Cut distributed processing compute costs by 15% using optimized PySpark workflows.
- Reliability & Quality:** Achieved 99.9% data accuracy with automated validation algorithms in healthcare pipelines.